The Journal of Open Source Education

ZERO-YOLO: A No-Code GUI for Training Custom Dataset on YOLO Segmentation Models

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DOI: 10.xxxxxx/draft

Software

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Submitted: 18 July 2025
Published: unpublished

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Summary

YOLO (YOLOv11/YOLOv8) segmentation models routinely achieve impressive scores on benchmark datasets (i.e. COCO dataset (Lin et al., 2014)), yet deploying them on a *new* dataset still demands (i) command-line / coding proficiency, and (ii) careful data re-formatting that many domain scientists lack. Commercial "auto-ML" platforms (e.g. Roboflow) (*Roboflow*, n.d.) simplify the process but require cloud upload of your own (possibly sensitive) image dataset and frequently impose paywalls or restrictive licences.

ZERO-YOLO closes this usability gap. It bundles the complete YOLO segmentation training pipeline into a single, Dockerised web application built with *Streamlit*. Researchers from biology, medicine, materials science, or any other field can—in a few clicks—convert local image–mask pairs into YOLO format, fine-tune a pretrained model, monitor training metrics live, and visualise predictions, all while their data never leave the workstation.

Statement of Need

Image segmentation is the process of dividing an image into distinct regions that correspond to different objects, structures, or classes of interest. It is a fundamental task in computer vision and image analysis, with critical applications across a wide range of fields.:

- **In healthcare**, segmentation helps radiologists delineate tumors, organs, or lesions in medical scans.
- **In agriculture**, researchers use segmentation to analyze crop health or identify pests from aerial imagery.
- **In environmental science**, ecologists segment satellite or drone images to study land use, habitats, or pollution.
- **In manufacturing**, engineers use segmentation for quality control and defect detection.
- **In education and research**, segmentation is vital for teaching and developing new computer vision techniques.

A YOLO segmentation model extends the "You Only Look Once" detector to predict pixel-precise masks and class labels in a single, end-to-end pass. Its lightning-fast inference and solid accuracy make it a go-to choice for real-time, resource-efficient vision tasks. While it shows amazing performance on benchmark datasets, *domain specialists* still face two practical barriers:

1. **Workflow complexity.** Preparing config files, writing training scripts, and troubleshooting CUDA/PyTorch errors deter non-ML experts (Jocher & others, 2024).

2. **Data sovereignty.** Institutions handling medical or proprietary imagery cannot upload data to third-party services (*Roboflow*, n.d.).

ZERO-YOLO addresses both. It eliminates coding entirely and keeps the full workflow on-premise under a permissive MIT licence, empowering scientists to run high-quality object-detection experiments without cloud dependencies or legal entanglements. The installation and usage instructions (both written and video) for the software can be found at: https://github.com/sumit-ai-ml/ZERO-YOLO

Features

ZERO-YOLO enables the full YOLO segmentation pipeline within a single unified GUI. Key features include:

- **Data Preparation & Harmonization**
  - Converts images and masks from `.png`, `.jpg`, `.jpeg`, `.tif`, `.tiff` into standardized `.tiff` format.
  - Ensures masks use correct pixel values per class and filenames are synchronized.
- **Automated Dataset Splitting**
  - Customizable train/validation/test splits (default 70/15/15).
- **YOLO-Compatible Annotation Conversion**
  - Segmentation masks auto-converted to YOLO `.txt` annotations.
  - Configurable class mapping via GUI.
- **YAML Configuration Generation**
  - Automated creation of YOLO YAML config files.
  - Upload class labels from Excel.
  - Adjust augmentation and hyperparameters: rotation, scaling, mosaic, mix-up, etc.
- **YOLO Training Pipeline**
  - Direct integration with Ultralytics YOLOv8 or any other advanced version of YOLO.
  - Configurable epochs, batch size, image size, device (GPU/CPU), layer freezing.
- **Interactive Visualization Playground**
  - Real-time visualization: original images, predicted masks, overlays, prediction confidence.
- **Mask Generation & Evaluation**
  - Automated inference to generate masks from new images.
  - Built-in metrics: Dice Score, Intersection-over-Union (IoU), sensitivity, specificity.

Software Description

Aspect	Detail
Language / stack	Python 3.10; Streamlit UI; PyTorch backend; Ultralytics YOLO engine
Distribution	One-command Docker Compose (CPU or NVIDIA GPU profile)

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Aspect	Detail
Input	Image folder + segmentation-mask folder + *label_names.xlsx* listing classes
Output	Trained `*.pt` weights, live training plots, interactive mask overlays
Licence	MIT

Internally the app launches a YOLO training process with user-selected epochs and model size, streams logs to the UI, and writes artefacts to the `runs/` directory. A small custom wrapper auto-generates YOLO YAML configs and converts segmentation masks to COCO-style polygons when needed.

Domain Impact

ZERO-YOLO democratizes access to advanced segmentation, benefiting a wide spectrum of domains:

- **Healthcare:** Medical image segmentation (e.g., CT, MRI) without coding.
- **Agriculture:** Crop monitoring, pest detection from field/drone imagery.
- **Environmental Science:** Segmentation of satellite/drone imagery for ecological studies.
- **Education:** Enables hands-on teaching of deep-learning segmentation without programming barriers.

References

Jocher, G., & others. (2024). *Ultralytics YOLOv8: Official documentation.*

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